

MicroAttention: Pre-Attention Token Compression for Efficient Transformers

Problem: Standard Transformer self-attention scales quadratically with sequence length: $O(n^2 \cdot d)$. As token count increases, compute and memory costs become a major bottleneck for long-context AI systems.

Core Idea: Before global Transformer attention, compress tokens locally using block-wise learned importance scoring. Instead of sending all n tokens into attention, partition them into blocks of size k and reduce each block into one compressed representation.

Mechanism:

1. Input Projection → Convert raw input into embeddings
2. Block Partition → Divide sequence into windows of size k
3. Token Scoring → Lightweight linear layer assigns token importance
4. Softmax Weighting → Normalize importance inside each block
5. Compression → Weighted sum produces one token per block
6. Reduced Transformer → Global reasoning over compressed sequence

Formula:

Original tokens = n

Compressed tokens = n/k

Attention cost reduces from $O(n^2)$ to $O((n/k)^2)$, plus lightweight compression cost $O(n)$.

Example:

$n = 100, k = 5 \rightarrow 100$ tokens become 20 tokens

Standard attention interactions: 10,000

Compressed interactions: 400 (~25x reduction)

Prototype Result (Sinwave Benchmark):

Block size 1 → Loss ≈ 0.0005

Block size 5 → Loss ≈ 0.0010

This suggests large compute savings with modest performance degradation on structured sequences.

Strengths:

- Major token reduction before expensive reasoning
- Learnable compression instead of fixed pooling
- Potential use in logs, sensor streams, long repetitive contexts, and edge AI

Limitations:

- Risk of losing rare critical tokens
- Fixed block sizes may break semantic boundaries
- May underperform on code, logic, or precision-heavy language tasks

Future Directions:

Adaptive block size, entropy-aware compression, multi-head MicroAttention, hybrid retention (summary + critical token), and benchmarking against Linformer, Longformer, and token pruning systems.

Conclusion: MicroAttention explores a broader systems principle: sacrificing small amounts of accuracy may be worthwhile if it dramatically reduces computational cost. If validated on complex tasks, it could become an efficiency-focused Transformer optimization strategy.